BÀI TẬP CHƯƠNG 3

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1. Tối ưu luận lý bằng phương pháp đại số Boolean cho các biểu thức sau:

a) $F(A, B, C) = AB + A\overline{B}C + AB\overline{C}$

$$= AB(1 + \overline{C}) + A\overline{B}C$$

$$= AB + A\overline{B}C = A(B + \overline{B}C) = A(B + C) = AB + AC$$

b) $F(X, Y, Z) = (X + Y)(X + \overline{Y})(X + Y + Z)$

$$f = (X + Y)(X + \overline{Y}) = X + X\overline{Y} + YX = X$$

c) $F(A, B, C, D) = A\overline{B}C + \overline{A}B + \overline{A}C + CD$

$$F = C(A\overline{B} + \overline{A}) + CD + \overline{A}B$$

$$= C(\overline{A}(1 + \overline{AB}) + A\overline{B}) + CD + \overline{A}B$$

$$= C(\overline{A} + \overline{AB} + A\overline{B}) + CD + \overline{A}B$$

$$= C(\overline{A} + \overline{B}) + CD + \overline{A}B$$

$$= \overline{A}C + \overline{B}C + CD + \overline{A}B$$

d) $F(A, B, C, D) = (A + B + C)(\overline{A} + B)(B + C + D)$

$$= (B + \overline{A}C)(B + C + D)$$

$$= B + BC + BD + \overline{A}CB + \overline{A}C + \overline{A}CD$$

$$= B + \overline{AC}$$

2. Chứng minh bằng đại số các biểu thức sau:

a) $\overline{(AB + \overline{A}\overline{B})} = \overline{A}B + A\overline{B}$

$$VT = \overline{A.B}.(A + B)$$

$$= (\overline{A} + \overline{B})(A + B)$$

$$= \overline{A}B + A\overline{B} = VP\ (dpcm)$$

b) $AB + \overline{A}C = (A + C)(\overline{A} + B)$

$$VP = AB + \overline{A}C + CB$$

$$= AB + \overline{A}C + CB(\overline{A} + A)$$

$$= AB + \overline{A}C + \overline{A}CB + ACB$$

$$= AB(1 + C) + \overline{A}C(1 + B) = AB + \overline{A}C = VT(dpcm)$$

c) $\overline{AC + B\overline{C}} = \overline{A}C + \overline{B}\overline{C}$

$$VT = (\overline{A} + \overline{C}).(\overline{B} + C) = \overline{A}.\overline{B} + \overline{A}C + \overline{C}.\overline{B}$$

$$= \overline{A}C + \overline{B}\overline{C} + \overline{AB}(C + \overline{C})$$

$$= \overline{A}C + \overline{B}\overline{C} + \overline{ABC} + \overline{AB}C$$

$$= \overline{A}C(1 + \overline{B}) + \overline{B}\overline{C}(1 + \overline{A})$$

$$= \overline{A}C + \overline{B}\overline{C} = VP(dpcm)$$

- Dùng luật đồng thuận để nhân phân phối biến dư thừa cho $1 = (X + \overline{X})$

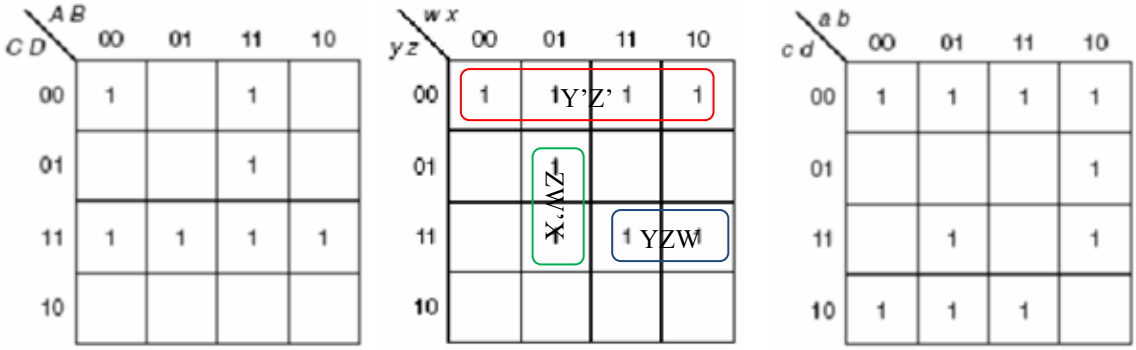
d) $(A + B)(\overline{A} + \overline{C})(B + \overline{C}) = (A + B)(\overline{A} + \overline{C})$

$$VT = (A + B)(\overline{A} + \overline{C})(B + \overline{C} + A\overline{A})$$

$$= (A + B)(\overline{A} + \overline{C})(B + \overline{C} + A)(B + \overline{C} + \overline{A})$$

$$= (A + B)(\overline{A} + \overline{C}) = VP(dpcm)$$

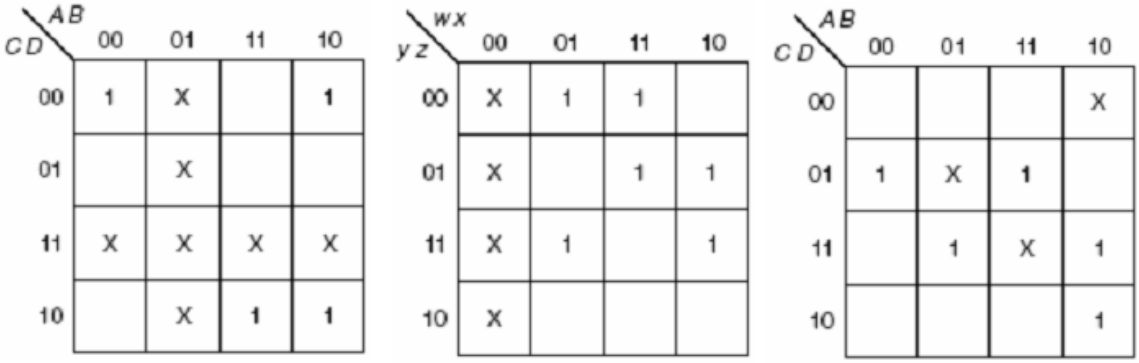
3. Đơn giản hóa các bìa Karnaugh sau:



a) $f(A, B, C, D) = CD + ABD + ABC + \overline{ABCD}$

b) $f(x, y, z, w) = \overline{yz} + \bar{x}wz + ywz$

c) $f(a, b, c, d) = \overline{c}\overline{d} + a\overline{b}d + \overline{a}c\overline{d} + bc\overline{d} + \overline{a}bc$



CD \ AB	00	01	11	10
00	1	X		1
01		X		
11	X	X	X	X
10		X	1	1

d)

yz \ wx	00	01	11	10
00	X	1	1	
01	X		1	1
11	X	1		1
10	X			

e)

CD \ AB	00	01	11	10
00				X
01	1	X	1	
11		1	X	1
10				1

f)

$$d)\ f(A,B,C,D) = BC + \overline{BCD} + A\overline{BD}$$

$$e)\ f(x,y,z,w) = x\overline{yz} + \overline{y}zw + \overline{x}zw + yz\overline{w}$$

$$f)\ f(A,B,C,D) = DB + \overline{A}.\overline{C}D + A\overline{B}C$$

4. Tối thiểu các biểu thức sau bằng phương pháp bìa-K
   a. F(X, Y, Z) = m1 + m2 + m3 + m4 + m6 + m7

$$F(X,Y,Z) = Y + \overline{X}Z + X.\overline{Z}$$

   b. G(X, Y, Z) = $M_0.M_2.M_3.M_5.M_7$

$$G(X,Y,Z) = (\overline{X} + Y).(X + Z).(\overline{X} + \overline{Z})$$

   c. H(A,B,C,D) = m0+m6+m8+m9 +m10 +m11+m13 +m14 +m15

$$H(A,B,C,D) = A\overline{B} + A\overline{C} + BC\overline{D} + \overline{BCD}$$

5. Tối thiểu các biểu thức sau bằng phương pháp bìa-K:
   a. F(x,y,z) = xy + xz' + yz + xyz

$$F(x,y,z) = x\overline{z} + yz$$

   b. G(a,b,c,d) = abc + ab'd + bc + a'bd + acd'

$$G(a,b,c,d) = bc + a\overline{b}d + \overline{a}bd + ac\overline{d}$$

   c. H(w,x,y,z) = (w' + x)(w+x+y).z'

$$H(x,y,z,w) = x\overline{z} + \overline{w}y\overline{z}$$

6. Tối thiểu các biểu thức sau theo dạng SoP hay PoS :
   a) $F(a, b, c, d) = \prod M(1, 7, 9, 13, 15)$

$$F = (\bar{b} + \bar{c} + \bar{d})(\bar{a} + \bar{b} + \bar{d})(\bar{a} + c + \bar{d})(b + c + \bar{d})$$

b) $F(w, x, y, z) = \sum m(0, 2, 5, 7, 8, 10, 13, 15)$

1			x
1	1		
1		x	
		1	

$$F = \bar{x}\bar{z} + xz$$

c) $F(w, x, y, z) = \sum m(0, 1, 3, 5, 14) + d(8, 15)$

$$F = wxy + \bar{w}\bar{x}z + \bar{w}\bar{y}z + \bar{x}\bar{y}\bar{z}$$

d) $F(a, b, c, d) = \prod M(1, 5, 9, 14, 15) \cdot D(11)$

$$F = (\bar{a} + \bar{b} + \bar{c})(a + c + \bar{d})(\bar{a} + b + \bar{d})$$